Network Traffic Analysis Report

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Incident Date: 4 Dec 2024

Report Date: 31 May 2026

Incident Reference: AgentTesla Exfiltration Event

Target Host: DESKTOP-VJCRXEB

Organization: Globex Manufacturing Ltd

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1. Executive Summary

On December 4, 2024, network traffic analysis confirmed an active AgentTesla malware infection on internal host **DESKTOP-VJCRXEB** (IP: 10[.]12][.]4[.]101). Following execution, the malware initiated an outbound DNS query to the legitimate domain api[.]ipify[.]org to perform external IP reconnaissance on the compromised host. The payload subsequently resolved the attacker-controlled domain ftp[.]ercolina-usa[.]com to external IP 192[.]254[.]225[.]136 and established an unencrypted FTP connection over Port 21. The malware then began a rapid multi-stage data exfiltration operation, uploading harvested browser credentials (PW_...), Google Chrome and Microsoft Edge session cookies (CO_...), and keylogger output (KL_...) to the attacker's FTP server. A 20-minute dormant collection phase was observed between 21:21:05 and 21:41:03, during which the keylogger module actively recorded victim keystrokes before a secondary FTP connection was re-established at 21:41:07.

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2. Infected Host Identification

2.1 Internal Victim Identification

- **Victim IP Address:** 10[.]12[.]4[.]101
- **Target Device:** Based on exfiltrated data filenames, the infected system belongs to user gary.strickman on host machine DESKTOP-VJCRXEB.

2.2 Host Behavior Indicators

The host focused its external communication heavily on a single remote destination, transmitting 87 packets out of a total 157-packet stream directly to external IP address 192[.]254[.]225[.]136. The system initiated an unauthorized, plain-text FTP (Port 21) connection to the external internet, bypassing standard corporate web browsing protocols. Immediately following authentication, the host executed rapid, automated file uploads (STOR commands), indicating programmatic malware control rather than normal human network activity. And this is an indication of an active information stealer malware.

2.3 First Signs of Compromise

The earliest indicator of system compromise was the initial automated network reconnaissance which occurred at 21:20:56 on 2024-12-04. The malware forced the host to query api[.]ipify[.]org. This was followed exactly 6 seconds later by the resolution of the attacker's FTP server, directly leading to the exfiltration sequence.

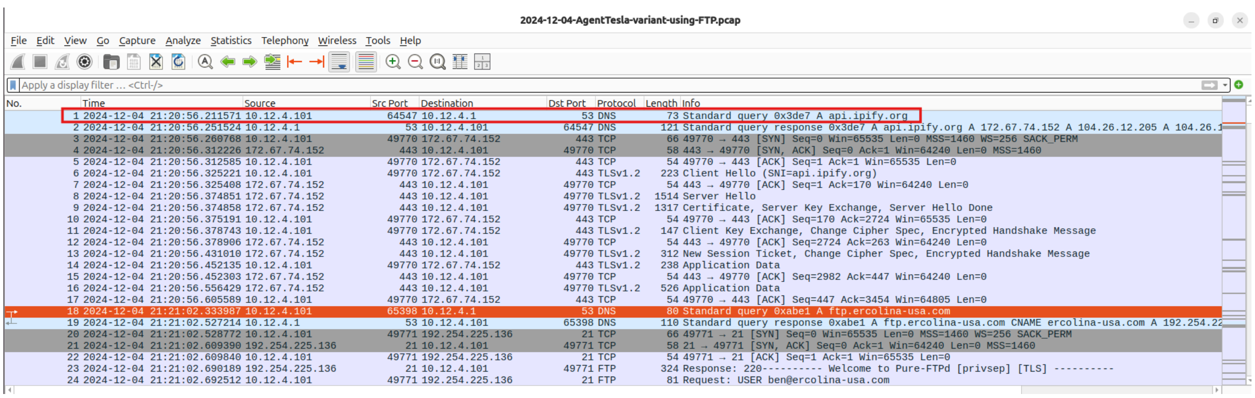

Figure 1: Wireshark showing victim IP and initial suspicious traffic

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3. DNS Analysis

1. **Suspicious Query:** ftp[.]ercolina-usa[.]com.The query was sent by the infected host to the internal DNS server (10[.]12[.]4[.]1), which returned a resolution to external IP address 192[.]254[.]225[.]136.
2. **Newly Observed Domains:** While the base domain ercolina-usa[.]com may be an older, legitimate site, the sub-domain ftp[.]ercolina-usa[.]com. is classified as a Newly Observed Domain relative to this local corporate network's baseline traffic.
3. **Malware-related resolution patterns:** The malware forces the system to look up api[.]ipify[.]org precisely 6 seconds prior to resolving the exfiltration server.

This pattern confirms that the malware checks the victim's external geographic routing profile before establishing an outbound connection to the C2 server. A second query to ftp[.]ercolina-usa[.]com. occurs at 21:41:07, exactly 20 minutes after the initial lookup. This periodic resolution pattern indicates automated persistence, allowing the malware to sleep, collect active keyboard inputs via its keylogger module, and check back in to drop off fresh logs.

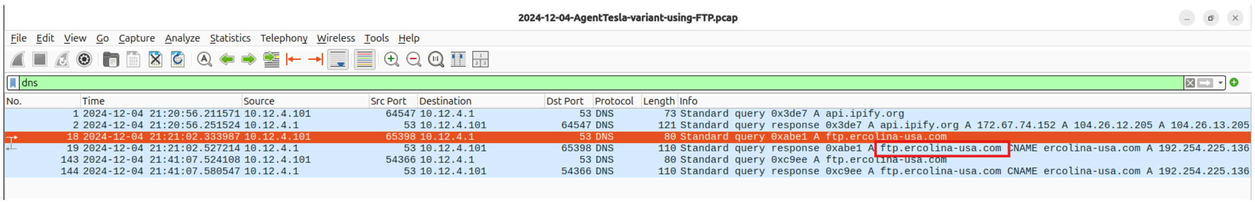

Figure 2: Wireshark DNS filter view showing suspicious queries and annotated newly observed domain

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4. FTP Traffic Analysis

Following the successful DNS resolution of the attacker-controlled domain, the infected host 10[.]12[.]4[.]101 initiated a TCP connection to the FTP server at 192[.]254[.]225[.]136 over Port 21. The FTP server responded with a 220 service ready banner, confirming the connection was successfully established. This marks the transition from the reconnaissance phase to the active exfiltration phase of the attack.

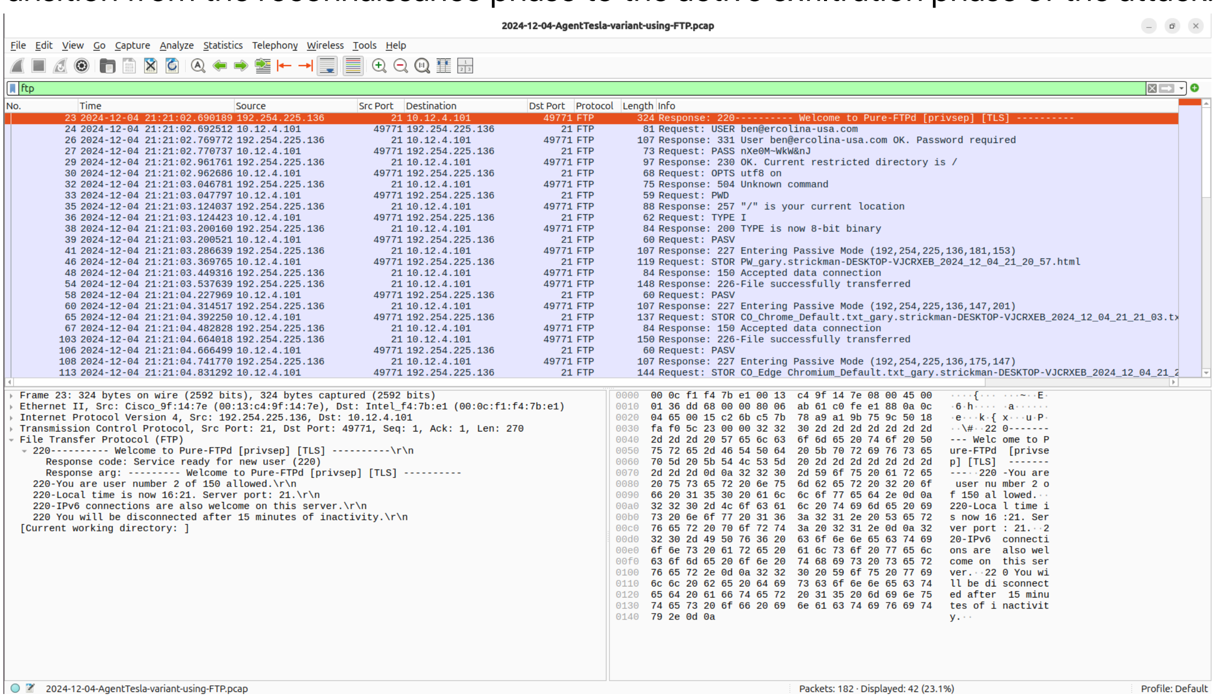

Figure 3: Wireshark display filter view showing FTP session initiation

4.1 Credentials Captured

- FTP Server IP Address: 192[.]254[.]225[.]136
- Port: 21
- Username: ben@ercolina-usa[.]com
- Password: nXe 0M~WkW&n

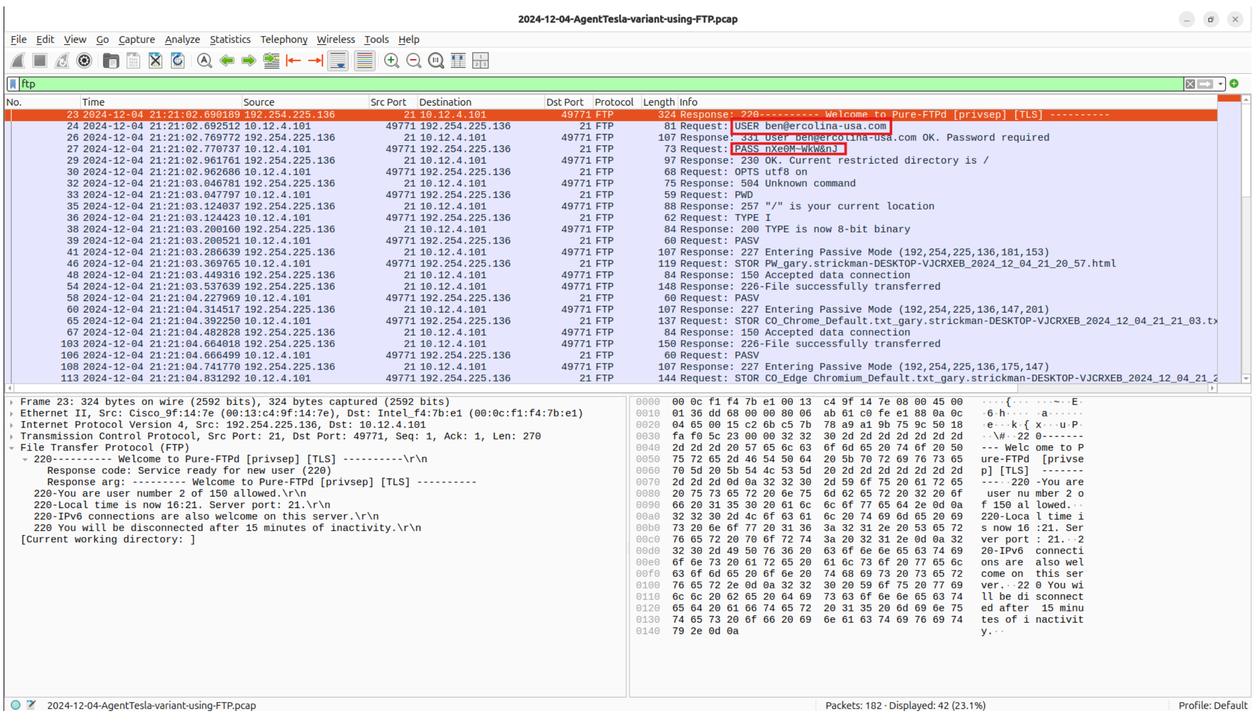

Figure 4: Wireshark FTP stream showing USER and PASS commands in cleartext

4.2 Data Exfiltration — STOR Commands

- File Uploads (STOR Commands) & Exfiltrated Names:
  - PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html
  - CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt
  - CO_Edge Chromium_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_04.txt
  - KL_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_41_04.html

4.3 Data Transfer Patterns:

- **The Sprint Phase**: The malware achieves a blistering, high-density initial data dump. It authenticates and uploads the password file (PW_) and both browser cookie files (CO_) in an ultra-fast 1-second burst between 21:21:03 and 21:21:04.
- **The Surveillance Gap**: The FTP channel goes entirely dormant for exactly 20 minutes and 4 seconds. During this network silence, the resident malware operates locally, recording user activity.
- **The Follow-Up Phase:** The malware wakes up, logs back in, and exfiltrates the keylogger file (KL_) at 21:41:08.

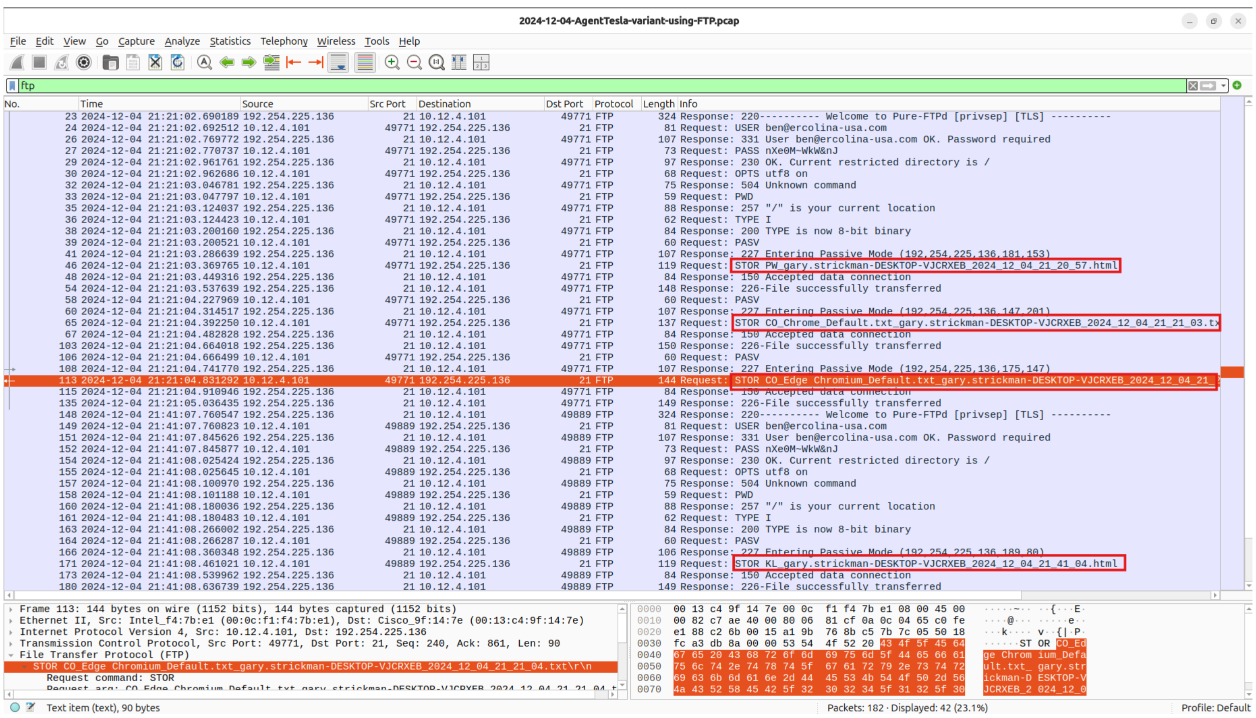

Figure 5: Wireshark showing STOR command and exfiltrated filename

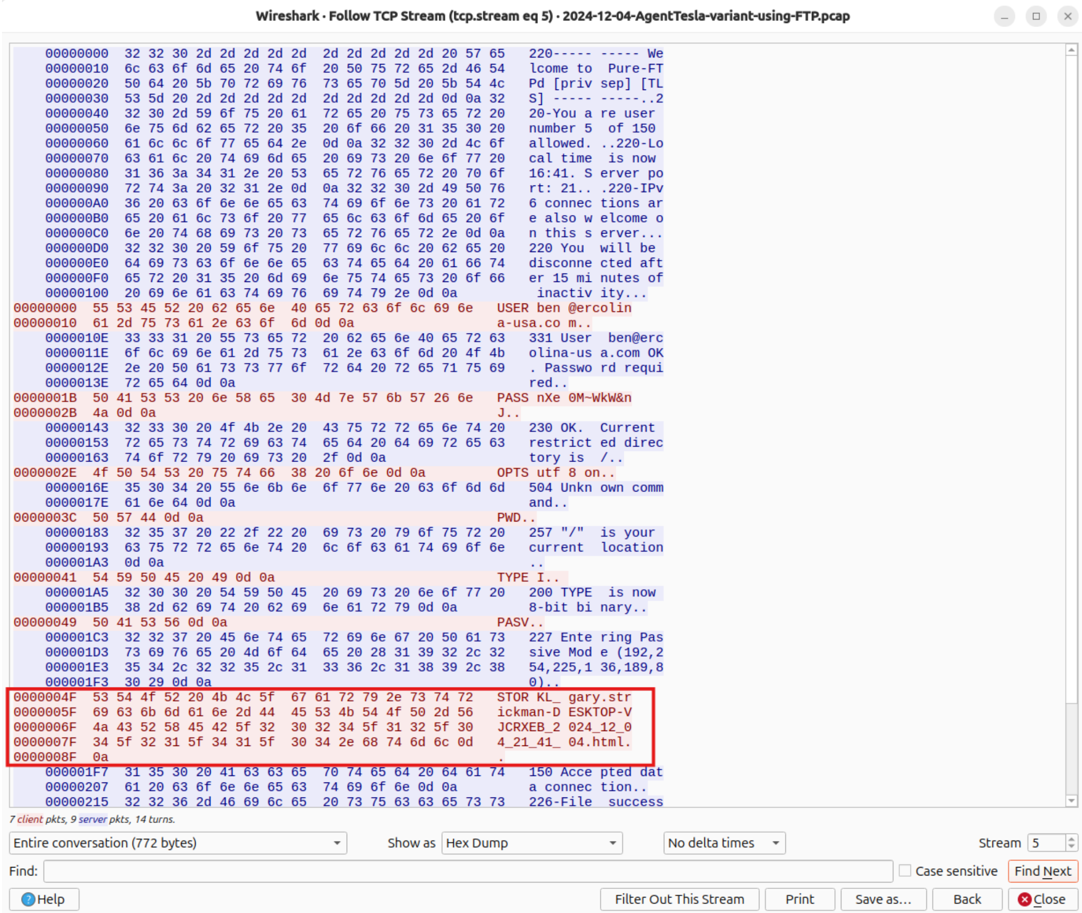

Figure 6: FTP data stream showing content of uploaded file

4.4 Summary of FTP Findings

- **Credentials stolen includes:** System application passwords and active web-browser session tokens harvested directly from Google Chrome and Microsoft Edge database directories.
- **File exfiltrated:** A password repository dump (PW_), two separate active browser session cookie logs (CO_), and a running live keystroke log capture (KL_).
- **Is FTP used for exfiltration by AgentTesla?** Yes. While AgentTesla holds capabilities to exfiltrate via SMTP or HTTP, this specific packet was explicitly hardcoded to use plain-text FTP outbound traffic to stream looted corporate assets to a remote staging database.

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5. Network-Based Indicators of Compromise (IOC) Table

This table details the explicit network markers.

IOC Type	Indicator Value	Threat Intelligence
Victim IP	10[.]12[.]4[.]101	Source host confirmed as the infected workstation, identified as the origin of all suspicious outbound DNS queries and FTP exfiltration traffic observed in the PCAP
FTP Server IP	192[.]254[.]225[.]136	External attacker-controlled FTP server resolved from ftp.ercolina-usa[.]com, serving as the primary data exfiltration destination for the AgentTesla payload
Domain Name	ftp[.]ercolina-usa[.]com	Attacker-controlled FTP domain resolved by the infected host at 21:21:02, used to establish the covert exfiltration channel for stolen credentials and browser session data
Reconnaissance Domain	api[.]ipify[.]org	Legitimate public IP lookup service abused by the AgentTesla payload during the initial execution phase to perform external IP reconnaissance on the victim host
FTP Username	ben@ercolina-usa[.]com	Plaintext FTP credential transmitted by the malware during the FTP login sequence, confirming attacker-controlled account access to the exfiltration server
FTP Password	nXe 0M~WkW&n	Plaintext FTP password captured in the cleartext FTP stream, confirming the absence of transport encryption and full credential exposure
Exfiltrated File - Credentials	PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html	Browser credential archive uploaded via FTP STOR command at 21:21:03, containing harvested saved passwords from the victim's browser profile
Exfiltrated File - Browsers Cookies	CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt	Browser session cookie archives for Google Chrome and Microsoft Edge uploaded via FTP STOR command, enabling potential session hijacking by the threat actor

	CO_Edge Chromium_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_04.txt	
Exfiltrated File - Keylog	KL_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_41_04.html	Keylogger output file uploaded via FTP STOR command at 21:41:08, containing recorded victim keystrokes captured by the AgentTesla keylogging module during the 20 minute dormant collection phase
Network Port	21 (Outbound)	Cleartext FTP control port used by the AgentTesla payload to establish an unencrypted command channel to the attacker-controlled exfiltration server, transmitting credentials and data with no transport layer security

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6. Full Attack Timeline

Figure 7: Full attack timeline reconstructed from PCAP, showing the complete AgentTesla attack chain from initial email delivery through data exfiltration and secondary beaconing on 04 December 2024

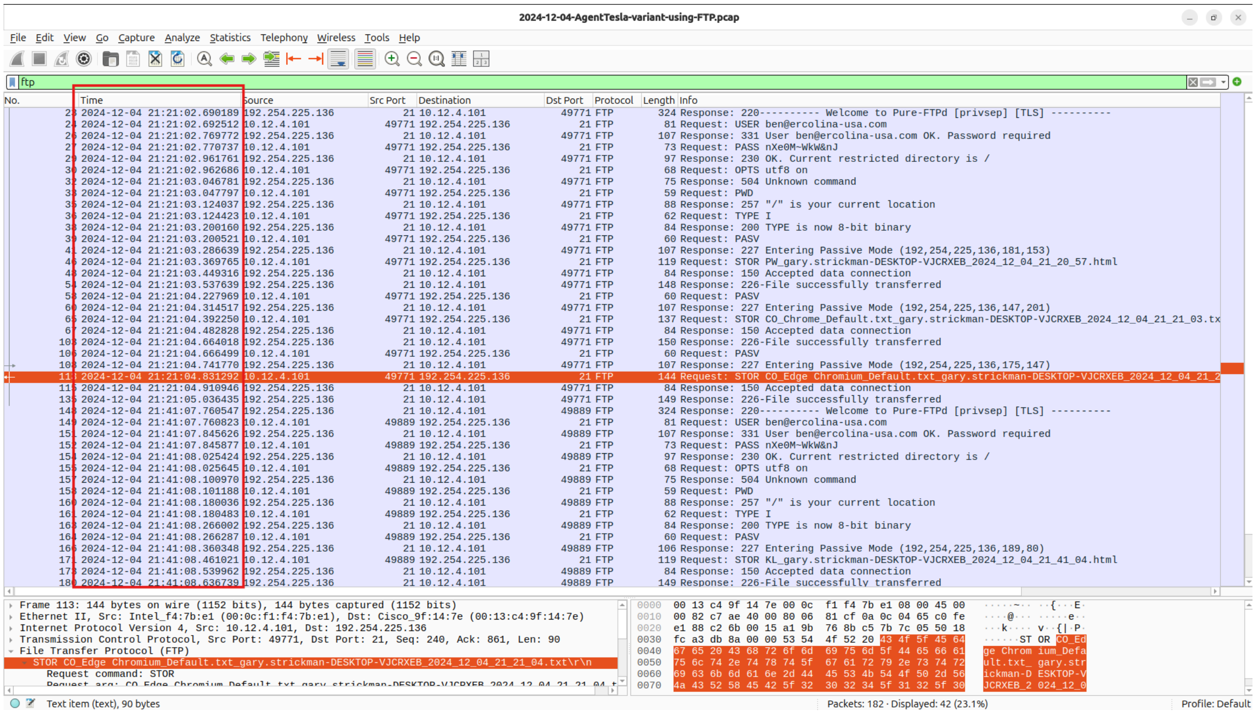

Figure 8: Wireshark timeline view showing full attack traffic sequence

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7. Defense Action

- Implement DNS filtering to block newly observed domains to prevent C2 domain resolution.
- Deploy an Endpoint Detection and Response solution on all hosts to detect future infostealer activity.
- Block all outbound FTP traffic permanently at the firewall to eliminate FTP exfiltration vector.
- Enable MFA on all corporate accounts to mitigate harvested credential impact.
- Conduct user awareness training focused on phishing to reduce human vulnerability.